Joseph Samuel Canzano

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EDUCATION

Ph.D.	**University of California, Santa Barbara** GPA: 3.81	**Dynamical Neuroscience**	Est. 2026
M.S.	**University of Florida** Concentration: Neuroscience GPA: 3.95	**Medical Science**	2019
B.S.	**University of North Florida** Concentration: Molecular/Cell Biology & Biotechnology Minor: Chemistry	**Biology**	2015

EMPLOYMENT & RESEARCH EXPERIENCE

2019-	**Graduate Student Researcher** PI: Spencer Smith	UCSB ECE Dept.
2017-2019	**Graduate Research Asst.** PI: Dr. Karim Oweiss	University of Florida ECE Dept.
2016-2017	**Biological Scientist I** PI: Dr. Martha Campbell-Thompson	University of Florida Experimental Pathology
2013-2015	**Undergraduate Research Asst.** PI: Dr. Evette Radisky	Mayo Clinic Cancer Biology
2011-2012	**Undergraduate Research Asst.** PI: Dr. John Hatle	University of North Florida Biology Dept.

PUBLICATIONS

2026 Rajpal, H., Stefens, C., Saeedian, M., **Canzano, J.S.**, Kareithi, M.G., Barahona, M., Smith, S.L., Schultz, S.R., Jensen, H.J. “Synergy mediates Long-Range Correlations in the Visual Cortex Near Criticality” **Front. Comput. Neurosci.**

2025 *Schneider, M., ***Canzano, J.S.**, *Peng, J., *Hou, Y., Smith, S.L., Beyeler, M. "Mouse vs. ai: A neuroethological benchmark for visual robustness and neural alignment" **ArXiv**

2025 Yu, C.H., Yu, Y., **Canzano, J.S.**, Fei, Y., Smith, S.L. "Non-inertial scan angle multiplier for expanded fields-of-view" **Biorxiv**

2023 Antoniades, A., Yu, Y., **Canzano, J.S.**, Wang, W., Smith, S.L. "Neuroformer: Multimodal and Multitask Generative Pretraining for Brain Data" **Arxiv**

2020 **Canzano J.S.**, Subramanian N, Castro R, Siddiqi A, Oweiss K, "In vivo two-photon imaging and parasympathetic neuromodulation of pancreatic microvascular dynamics in rats" **Biorxiv**

2018 **Canzano, J.,** Nasif L.H., Butterworth E.A., Fu, A., Atkinson, M., Campbell-Thompson, M. "Islet Microvasculature Alterations with Loss of Beta-Cells in Patients with Type 1 Diabetes." **J Histochem Cytochem** [PMID: 29771178]

2014 Tokar, D.R., Veleta, K.A., **Canzano J.**, Hahn, D.A., Hatle, J.D. "Vitellogenin RNAi Halts Ovarian Growth and Diverts Reproductive Proteins and Lipids in Young Grasshoppers." **Integr. Comp. Biol.** [PMID: 24920749]

TALKS/PRESENTATIONS

2025 Workshop *M Schneider, ***JS Canzano**, *J Peng, *Y Hou, SL Smith, M Beyeler "Mouse vs. AI: A Neuroethological Benchmark for Visual Robustness" **NeurIPS 2025**

2025 Poster H Rajpal, C Stefens, M Saeedian, **JS Canzano**, M Barahona, S Schultz, SL Smith, HJ Jensen, "Synergistic information processing as the mechanism for long-range correlations in the brain." **Network Neuroscience 2025**

2024 **Invited Talk** **JS Canzano**, M Duron, SL Smith, "Large scale population coding during open-world behavior guided by complex visual stimuli" **Institute of Science, Tokyo.** Host: Riichiro Hira

2024 **Talk** **JS Canzano**, M Duron, SL Smith, "Large scale population coding during open-world behavior guided by complex visual stimuli" **Japan Neuroscience Society Conference 2024**

2024 Poster C Stefens, H Rajpal, **JS Canzano**, M Saeedian, M Yang, LD Arcangelis, HJ Jensen, M Barahona, SL Smith, SR Schultz,

		"Mesoscopic multiphoton calcium imaging reveals a confluence of overlapping avalanches with varying distance to criticality and distinct roles" **SfN 2024**
2024	Poster	A Antoniades, Y Yu, **JS Canzano**, WY Wang, SL Smith, "Neuroformer: Multimodal and Multitask Generative Pretraining for Brain Data" **ICLR 2024**
2023	Poster	**JS Canzano**, M Duron, SL Smith, "Navigation and high-level visual features in mouse posterior cortex", UCSB ECE Graduate Summit
2019	Poster	N. SUBRAMANIAN, ***JS CANZANO**, B. GOOLSBY, H. KIM, P. NAVARRO, R. CASTRO, K. G. OWEISS "Simultaneous two-photon calcium imaging and cell-targeted optogenetic stimulation shapes somatosensory cortex plasticity" **SfN 2019, *first author** placed second for registration technicality
2019	Poster	B GOOLSBY, HJ KIM, **JS CANZANO**, K. G. OWEISS "All-optical functional mapping of association cortex subserving working memory and perceptual decision making" **SfN 2019**
2019	**Talk**, Poster	**JS Canzano**, N Subramanian, A Siddiqi, R Castro, R Johnson, K Oweiss, "In Vivo Two-Photon Imaging of Pancreatic Sympathetic and Parasympathetic Microvessel Neuromodulation" **11th Congress of the International Society for Autonomic Neuroscience**
2019	Poster	**JS Canzano**, H Kim, R Castro, K Oweiss "Artificial Forelimb Somatosensation via Optogenetic Stimulation and Two-Photon Calcium Imaging in a Sensory-Guided Motor Task" **Sculpted Light in the Brain Conference**
2019	Poster	**JS Canzano**, R Castro, H Kim, K Oweiss, "Artificial Forelimb Somatosensation via Optogenetic Stimulation and Two-Photon Calcium Imaging in a Sensory-Guided Motor Task", **Max Planck Florida Institute Sunposium 2019**
2017	**Talk**, Poster	"Endocrine and Exocrine Microvascular Alterations in Human Type 1 Diabetes," **JDRF nPOD Annual Scientific Meeting**
2016	Presentation	"Endocrine and Exocrine Microvascular Alterations in Human Type 1 Diabetes," **UF Diabetes Institute "Works in Progress"**

2013	Presentation	“Understanding mesotrypsin: The role of residue 193 in the unusual reverse activity of human mesotrypsin and similar vertebrate enzymes,” **UNF Mayo Clinic Intern Talks**
2012	Poster	“Vitellogenin RNAi halts ovarian growth and diverts reproductive proteins and lipids in young grasshoppers,” **UNF SOARS Conference**
2012	Poster	“Physiological effects of RNAi for vitellogenin on reproduction and fat body storage in grasshoppers,” **Society for Integrative Comparative Biology Annual Meeting**

TEACHING

TA positions:

- **25% TA, PSY 115 Neuropharmacology**, UCSB, Instructor: Karen Szumlinski
- **50% TA, PSY 106 Intro Biopsych,** UCSB. Instructor: Samantha Scudder
- **50% TA, PSY 111 Behavioral Neuroscience**, UCSB. Instructor: Vanessa Woods
- Reader, **PSY161 Systems Neuroscience**, UCSB. Instructor: Michael Goard

Training:

- UCSB PBS TA training course series

MENTORSHIP

Undergraduate mentees:

- Vivian Lin, animal behavior, 2025-present
- Janelle Rader, animal behavior, 2022-2025
- Miranda Duron, animal behavior, 2020-2022
- Abdurahman Siddiqi, peripheral neuromodulation, 2018-2019
- Lith Nasif, peripheral neural immunofluorescence, 2016-2017

SELECTED ACADEMIC COURSEWORK

Mo/Cell Bio & Neurobio   Systems, Dynamics, Machine Learning   Chemistry & Physics

Graduate	Undergraduate
Computational Neuroscience (UCSB)	Molecular/Cell Biology
Machine Learning (UCSB)	Genetics
Linear Systems (UCSB)	Biochemistry
Systems Neuroscience (UCSB)	Physiology
ODE's (systems of) (UCSB)	Virology
Biological dynamics (UCSB)	Computer Science
Measuring/Manipulating Neural Activity (UCSB)	Multivariate Calculus
Neuropharmacology (UF)	Physical Chemistry
Molecular/Cell Neuroscience (UF)	Inorganic Chemistry
Neural Signals, Systems & Tech. (UF)	Psychobiology
Functional Human Neuroanatomy (UF)	Statistics
Fund. of Biomedical Science (UF)	

WORKSHOP PARTICIPATION

2022 **Neuropixels and OpenScope Workshop**, Allen Institute, **2022**

PROFESSIONAL MEMBERSHIPS

2024- Japanese Neuroscience Society
2019- Society for Neuroscience

AWARDS AND HONORS

2019 **Travel award, Sculpted Light in the Brain 2019**
2013 Awarded joint UNF Biology - Mayo Clinic internship

REFERENCES

Available on request